

The Parameter-Derivative Tower of the Generalized Stieltjes Constants: a Uniform Harmonic-Number Theorem

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Abstract

The generalized Stieltjes constants $\gamma_n(a)$, regarded as functions of the Hurwitz parameter a , have a parameter-derivative tower whose k -th rung lives at the $s = k + 1$ derivative layer of the Hurwitz zeta function. We show this tower is *uniform in k* and governed entirely by the harmonic numbers: writing $H_k = \sum_{m \leq k} 1/m$ and $H_k^{(2)} = \sum_{m \leq k} 1/m^2$,

$$\gamma_1^{(k)}(a) = (-1)^{k+1} k! [\zeta'(k+1, a) + H_k \zeta(k+1, a)],$$

$$\gamma_2^{(k)}(a) = (-1)^k k! [\zeta''(k+1, a) + 2H_k \zeta'(k+1, a) + (H_k^2 - H_k^{(2)}) \zeta(k+1, a)],$$

and likewise for every n , with the elementary symmetric functions of $\{1, \frac{1}{2}, \dots, \frac{1}{k}\}$ as coefficients. The same H_k controls the rational-argument landscape: a character coordinate $\sum_p \chi(p) \gamma_1^{(k)}(p/q) = (-1)^{k+1} k! q^{k+1} [L'(k+1, \chi) + (H_k + \ln q) L(k+1, \chi)]$, a distribution row, and a residual-lattice dimension $\varphi(q) - 1$ that is *independent of k* . We give explicit closed-form tables for the second-constant derivatives $\gamma_2'(p/q)$ and $\gamma_2''(p/q)$ at $q \in \{1, 2, 3, 4, 6\}$, and we establish the derivative $\leftrightarrow$ negapolygamma duality at all orders: the functional equation pairs $s = k + 1$ with $s = -k$, the layer of the negative-order polygamma $\psi^{-(k+1)}$, through the harmonic-number bridge $L'(k+1, \chi)/L(k+1, \chi) + \overline{L'(-k, \chi)/L(-k, \chi)} = \gamma + \ln(2\pi/q) - H_k$, with a clean parity dichotomy. As a representative corollary, $\gamma_1''(\frac{1}{4}) = -256\pi^3 \psi^{(-3)}(\frac{1}{4}) + (\text{elementary})$.

1 Introduction and notation

The generalized (Hurwitz) Stieltjes constants $\gamma_n(a)$ are the Laurent coefficients of the Hurwitz zeta function at its pole,

$$\zeta(s, a) = \frac{1}{s-1} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(a) (s-1)^n, \quad (1)$$

so $\gamma_0(a) = -\psi(a)$ and $\gamma_n(1) = \gamma_n$ is the ordinary n -th Stieltjes constant. The word “constant” is a misnomer: each γ_n is a smooth function of a on $(0, \infty)$, and the object of this note is the *parameter-derivative tower*

$$\gamma_n^{(k)}(a) := \frac{d^k}{da^k} \gamma_n(a), \quad k = 0, 1, 2, \dots$$

*v.reshetnikov@gmail.com. Prepared with agentic assistance (Anthropic Claude) in the *Smithereens* repository, github.com/VladimirReshetnikov/Smithereens, src/PolylLog. Provenance: created 2026-06-28 (UTC); repository snapshot f594924c0dcc98e0fcb8147b287a09aa52f75955. Every displayed identity was verified numerically (mpmath, 60–220 digits), the $q = 4$ entries independently in a second computer-algebra system (Wolfram, ~ 426 digits), and the whole development was re-derived by an independent multi-agent workflow.

Blagouchine [1] gave the closed-form theory of $\gamma_1(p/q)$ at rational arguments; the present development is the analogue “one (and two) order(s) up,” for the derivatives in the parameter, which had not been tabulated.

Throughout, $\zeta^{(j)}(s_0, a) = \partial_s^j \zeta(s, a)|_{s=s_0}$ and $L^{(j)}(s_0, \chi)$ similarly for a Dirichlet L -function; $H_k = \sum_{m=1}^k 1/m$, $H_k^{(2)} = \sum_{m=1}^k 1/m^2$ (with $H_0 = H_0^{(2)} = 0$); $\psi^{(k)}$ is the k -th derivative of the digamma function and $\psi^{(-n)}$ the negative-order polygamma (Wolfram’s `PolyGamma`[- n , $\cdot$]); A is the Glaisher–Kinkelin constant, G Catalan’s constant, β the Dirichlet beta function, and χ_{-3}, χ_{-4} the odd quadratic characters of conductors 3, 4.

2 The uniform master identity

Theorem 2.1 (Harmonic-number master identity). *For all $k \geq 1$ and $n \geq 0$,*

$$\gamma_n^{(k)}(a) = (-1)^{n+k} k! n! \sum_{i+j=n} e_i(k) \frac{\zeta^{(j)}(k+1, a)}{j!}, \quad (2)$$

where $e_i(k)$ is the i -th elementary symmetric function of $\{1, \frac{1}{2}, \dots, \frac{1}{k}\}$; in particular $e_0 = 1$, $e_1 = H_k$, $e_2 = \frac{1}{2}(H_k^2 - H_k^{(2)})$. The first two rows are

$$\gamma_1^{(k)}(a) = (-1)^{k+1} k! [\zeta'(k+1, a) + H_k \zeta(k+1, a)], \quad (3)$$

$$\gamma_2^{(k)}(a) = (-1)^k k! [\zeta''(k+1, a) + 2H_k \zeta'(k+1, a) + (H_k^2 - H_k^{(2)}) \zeta(k+1, a)]. \quad (4)$$

Proof. From $\partial_a \zeta(s, a) = -s \zeta(s+1, a)$, iteration gives $\partial_a^k \zeta(s, a) = (-1)^k (s)_k \zeta(s+k, a)$ with the rising factorial $(s)_k = s(s+1) \cdots (s+k-1)$. For $k \geq 1$ this annihilates the pole term in (1), so

$$\sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n^{(k)}(a) (s-1)^n = (-1)^k (s)_k \zeta(s+k, a).$$

Set $s = 1 + \varepsilon$. Then $(s)_k = (1 + \varepsilon)_k = \prod_{m=1}^k (m + \varepsilon) = k! \prod_{m=1}^k (1 + \varepsilon/m) = k! \sum_{i \geq 0} e_i(k) \varepsilon^i$, the generating identity for the elementary symmetric functions of $\{1/m\}$, while $\zeta(s+k, a) = \zeta(k+1+\varepsilon, a) = \sum_{j \geq 0} \zeta^{(j)}(k+1, a) \varepsilon^j / j!$ is analytic at $\varepsilon = 0$ because $k+1 \geq 2$. Comparing coefficients of ε^n yields (2); $n = 1, 2$ give (3)–(4). $\square$

Theorem 2.1 is the harmonic-number repackaging of Coffey’s all-orders Stirling-number ladder [2]: the change of basis from the unsigned Stirling numbers $|s(k+1, i+1)|$ to the symmetric functions $e_i(k)$ turns the coefficients into $H_k, H_k^2 - H_k^{(2)}, \dots$. The specializations $\gamma_1' = \zeta'(2, a) + \zeta(2, a)$, $\gamma_1'' = -2\zeta'(3, a) - 3\zeta(3, a)$, $\gamma_2' = -[\zeta''(2, a) + 2\zeta'(2, a)]$ and $\gamma_2'' = 2\zeta''(3, a) + 6\zeta'(3, a) + 2\zeta(3, a)$ are the $s = 2, 3$ layers used below. Identity (2) was verified against numerical k -th parameter-derivatives of γ_1, γ_2 for $k = 1, \dots, 5$.

3 The uniform rational-argument landscape

Three structural facts hold uniformly in k ; all follow from $\sum_p \chi(p) \zeta(s, p/q) = q^s L(s, \chi)$ (and its multiplicative cousin) differentiated in s , fed through Theorem 2.1.

Proposition 3.1 (Character coordinate). *For a primitive Dirichlet character χ modulo q and all $k \geq 1$,*

$$\sum_{p=1}^{q-1} \chi(p) \gamma_1^{(k)}(p/q) = (-1)^{k+1} k! q^{k+1} [L'(k+1, \chi) + (H_k + \ln q) L(k+1, \chi)]. \quad (5)$$

Proposition 3.2 (Distribution row). *For $d \geq 2$ and all $k \geq 1$, with $\psi^{(k)}(a) = (-1)^{k+1} k! \zeta(k+1, a)$,*

$$\sum_{j=0}^{d-1} \gamma_1^{(k)}\left(\frac{a+j}{d}\right) = d^{k+1} \gamma_1^{(k)}(a) + d^{k+1} \ln d \psi^{(k)}(a). \quad (6)$$

Proof of (5) and (6). Apply $\sum_p \chi(p) (\cdot)$ (resp. $\sum_{j < d} (\cdot)$) to (3). For (5), use $\sum_p \chi(p) \zeta(s, p/q) = q^s L(s, \chi)$ and its s -derivative $q^s \ln q L(s, \chi) + q^s L'(s, \chi)$ at $s = k+1$; collect. For (6), use $\sum_{j < d} \zeta(s, (a+j)/d) = d^s \zeta(s, a)$ and its s -derivative, and recognize $(-1)^{k+1} k! \zeta(k+1, a) = \psi^{(k)}(a)$. $\square$

Theorem 3.3 (k -independent rank). *Fix a grid denominator q . The $\mathbb{Q}$ -vector space of linear relations among $\{\gamma_1^{(k)}(r/q) : 1 \leq r < q\}$ generated by the distribution rows (6) has, for every $k \geq 1$, residual dimension*

$$(q-1) - \text{rank} = \varphi(q) - 1,$$

one residual direction per non-principal character mod q , both parities.

The rank was computed by exact rational row reduction; for $k = 1, 2, 3$ and every $q = 3, \dots, 30$ the residual vectors are bitwise identical (the collapsed coordinate carries weight d^{k+1} , which changes the integer entries but not the rank). Thus differentiation in the parameter destroys, at every order, the Malmsten–Blagouchine reflection halving $\varphi(q)/2 - 1$ of the underived constants. Equation (5) was verified for $\chi_{-3}, \chi_{-4}, \chi_5$ and $k = 1, \dots, 5$; (6) for $k = 1, \dots, 5$, $d \in \{2, 3, 5\}$.

4 Explicit tables for the second constant

We record the closed forms of $\gamma_2' = -[\zeta''(2, a) + 2\zeta'(2, a)]$ (the $s = 2$ layer) and $\gamma_2'' = 2\zeta''(3, a) + 6\zeta'(3, a) + 2\zeta(3, a)$ (the $s = 3$ layer) at the elementary denominators. The new constants entering are the *second* s -derivatives $\zeta''(2), \zeta''(3), \beta''(s) = L''(s, \chi_{-4})$ and $L''(s, \chi_{-3})$; the values $L(3, \chi_{-4}) = \pi^3/32$ and $L(3, \chi_{-3}) = 4\pi^3/(81\sqrt{3})$ are elementary. As at the first-derivative layers, the odd-character block flips sign as a single unit under $p \mapsto q - p$ (including any elementary π -power value-part it carries). All entries are exact and were verified to $< 10^{-70}$.

Second constant, first parameter-derivative.

$$\begin{aligned} \gamma_2'(1) &= -\zeta''(2) - 2\zeta'(2), \\ \gamma_2'(\tfrac{1}{2}) &= -3\zeta''(2) - (6 + 8 \ln 2)\zeta'(2) - \tfrac{4}{3}\pi^2 \ln 2 - \tfrac{2}{3}\pi^2 \ln^2 2, \\ \gamma_2'(\tfrac{1}{3}) &= -4\zeta''(2) - (8 + 9 \ln 3)\zeta'(2) - \tfrac{3}{2}\pi^2 \ln 3 - \tfrac{3}{4}\pi^2 \ln^2 3 \\ &\quad - \left[\tfrac{9}{2}L''(2, \chi_{-3}) + 9(1 + \ln 3)L'(2, \chi_{-3}) + (9 \ln 3 + \tfrac{9}{2} \ln^2 3)L(2, \chi_{-3}) \right], \\ \gamma_2'(\tfrac{1}{4}) &= -6\zeta''(2) - (12 + 28 \ln 2)\zeta'(2) - \tfrac{14}{3}\pi^2 \ln 2 - 5\pi^2 \ln^2 2 \\ &\quad - \left[8\beta''(2) + 16(1 + 2 \ln 2)\beta'(2) + (32 \ln 2 + 32 \ln^2 2)\mathbf{G} \right], \end{aligned}$$

with $\gamma'_2(2/3), \gamma'_2(3/4)$ obtained by negating the bracketed odd-character block, and the $q = 6$ entries given in the source report.

Second constant, second parameter-derivative.

$$\begin{aligned}
\gamma''_2(1) &= 2\zeta''(3) + 6\zeta'(3) + 2\zeta(3), \\
\gamma''_2(\tfrac{1}{2}) &= 14\zeta''(3) + (42 + 32\ln 2)\zeta'(3) + (14 + 48\ln 2 + 16\ln^2 2)\zeta(3), \\
\gamma''_2(\tfrac{1}{3}) &= 26\zeta''(3) + (78 + 54\ln 3)\zeta'(3) + (26 + 81\ln 3 + 27\ln^2 3)\zeta(3) \\
&\quad + \left[27L''(3, \chi_{-3}) + (81 + 54\ln 3)L'(3, \chi_{-3}) + (27 + 81\ln 3 + 27\ln^2 3)L(3, \chi_{-3}) \right], \\
\gamma''_2(\tfrac{1}{4}) &= 56\zeta''(3) + (168 + 240\ln 2)\zeta'(3) + (56 + 360\ln 2 + 248\ln^2 2)\zeta(3) \\
&\quad + \left[64\beta''(3) + (192 + 256\ln 2)\beta'(3) + (2 + 12\ln 2 + 8\ln^2 2)\pi^3 \right],
\end{aligned}$$

again with $\gamma''_2(2/3), \gamma''_2(3/4)$ the odd-block negations. The reducible fractions collapse correctly ($\frac{2}{4} = \frac{3}{6} = \frac{1}{2}$, $\frac{2}{6} = \frac{1}{3}$), an internal consistency check.

5 The derivative $\leftrightarrow$ negapolygamma duality

The functional equation pairs the layer $s = k + 1$ of $\gamma_1^{(k)}$ with the layer $s = -k$, which is exactly where the negative-order polygamma $\psi^{(-k+1)}$ lives (its ladder uses $\zeta'(-k, \cdot)$). The seam is a single uniform identity.

Theorem 5.1 (Harmonic-number bridge law). *Let χ be a primitive character mod q for which $L(-k, \chi) \neq 0$. Then*

$$\frac{L'(k+1, \chi)}{L(k+1, \chi)} + \overline{\left(\frac{L'(-k, \chi)}{L(-k, \chi)} \right)} = \gamma + \ln \frac{2\pi}{q} - H_k. \quad (7)$$

The same H_k that sets the master coefficient in Theorem 2.1 sets the bridge constant; the case $k = 1$ recovers the generalized-Glaisher constant $C_q = \gamma + \ln(2\pi/q) - 1$.

Remark 5.2 (Parity dichotomy). Since $L(-k, \chi) = 0$ if and only if $\chi(-1) = (-1)^k$, the direct form (7) applies precisely when $\chi(-1) \neq (-1)^k$; otherwise the $s = -k$ side has a trivial zero and (7) holds in its first-nonzero-order variant (a ratio of leading nonzero derivatives). This is what makes the *value-level* corollary parity-selective.

At $z = \frac{1}{4}$ the governing character is χ_{-4} (odd). For k even there is no trivial zero, and $\psi^{(-(k+1))}(\frac{1}{4})$ carries the genuine derivative atom $\beta'(k+1) = L'(k+1, \chi_{-4})$, so $\gamma_1^{(k)}(\frac{1}{4})$ is an exact finite combination of $\psi^{(-(k+1))}(\frac{1}{4})$ and elementary constants. The representative instance $k = 2$ is, after eliminating $\beta'(3)$ entirely,

$$\begin{aligned}
\gamma''_1(\tfrac{1}{4}) &= -256\pi^3 \psi^{(-3)}(\tfrac{1}{4}) - 56\zeta'(3) - 84\zeta(3) - 120\ln 2 \zeta(3) + 35\pi \zeta(3) \\
&\quad - 3\pi^3 - 4\ln 2 \pi^3 + 64\ln A \pi^3 - 2\gamma \pi^3 + 2\ln(2\pi) \pi^3,
\end{aligned} \quad (8)$$

verified to $< 10^{-118}$. (The irrational coefficient $-256\pi^3$ on $\psi^{(-3)}(\frac{1}{4})$ is why a rational integer-relation search for this identity must take $\pi^3\psi^{(-3)}$, not $\psi^{(-3)}$, as the search atom.) For k odd the value-level duality *degenerates*: by Remark 5.2, $\psi^{(-(k+1))}(\frac{1}{4})$ then carries only the lower *value* atom (for $k = 1$, $\beta'(-1) = 2G/\pi$, elementary; for $k = 3$, $\beta'(-3) \sim \beta(4)/\pi^3$), while the genuine

derivative atom $\beta'(k+1)$ of $\gamma_1^{(k)}(\frac{1}{4})$ is tied instead to the second derivative $\beta''(-k)$, absent from $\psi^{(-k+1)}(\frac{1}{4})$. Numerically the coefficient of $\beta'(k+1)$ in $\psi^{(-k+1)}(\frac{1}{4})$ is zero to $\sim 10^{-163}$ for $k = 1, 3$. The clean odd- k value-level duality therefore lives at even-character points (e.g. $q = 5$), where the parity is reversed.

6 The new atoms are genuine

The γ_2 tables of §4 introduce six second- s -derivative constants: $\zeta''(2), \zeta''(3), \beta''(2), \beta''(3), L''(2, \chi_{-3}), L''(3, \chi_{-3})$. A PSLQ sweep at 200 digits with planted controls (the elementary values $\beta(3) = \pi^3/32$, $L(3, \chi_{-3}) = 4\pi^3/(81\sqrt{3})$, $\zeta(2) = \pi^2/6$ are recovered; a random constant is rejected) finds *no* reduction of any of the six over a basket of the lower value and first-derivative atoms with $\gamma/\log$ products, and no relation among them: each is a genuine, mutually independent atom, the natural continuation of the $\beta'(2), \beta'(3)$ findings at the first-derivative layers. This was reproduced by an independent agent with its own controls.

Verification record

Every identity above was checked numerically in the *Smithereens* repository: the master identity, character coordinate, distribution row, rank theorem, γ_2 tables, bridge law and the duality (8) by the self-test battery in `src/PolyLog/tools/stieltjes_general_k.py` (with `scratch/g1k_grid_rank.py` for the exact rank and `stieltjes_s2deriv_hunts.py` for §6); the $q = 4$ table entries and the master identities additionally in an independent computer-algebra system. An independent multi-agent workflow re-derived the γ_2'' table (all entries to $< 10^{-128}$), re-verified the general- k claims at the independent point $a = \sqrt{2} - 1$ for $k = 1, \dots, 5$, confirmed the bridge law in both parities ($\leq 2.7 \times 10^{-161}$), and reproduced the atom sweep.

References

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